



KIASORT Instructions:

Overview

The GUI has six tabs; in this workshop, we will focus on **Main**, **Channels**, and **Curation**.

1. Main Tab

RUN Subtab

1. Select Input Data File

- Click the “Input Data File” field and choose your data file (one single .bin or .dat file for all channels).

2. Select Output Path

- Point this to any folder, typically, the same folder that contains your input data.

3. Select Channel Map File

- Navigate to and select your channel map file.

Denoising

- Set **Denoising** Check Box **True** if the data requires denoising. Check Box true if the data has extreme noise in the high frequency range.
- Remove shared noise between channels, adjust as needed depending on your data.
 - **Noisy Ch Percentile:** % of channels showing the noise pattern.

- **Noise Corr. Thr:** Correlation threshold to detect shared noise.
- **Whitening:** Whitens the data. Sometimes in high noise conditions you might need to check both box true.

Configurations

- Locate the “Configs” section and verify or adjust these settings:
 - **Channel selection radius:** Set the radius for selecting the number of channels before and after the main channels.
 - **Max Half Channel count:** Half maximum number of channels used for waveform extraction.
 - **Sampling Freq:** Ensure it is your data sampling frequency (**Hz**).
 - **Min Thr:** Typically, between **6–9**; you can select an appropriate value through **Channels Tab**.
 - All other parameters have sensible defaults. Feel free to adjust them if your data require different preprocessing or filtering.
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2. Channels Tab

1. Plot Channels

- Click **Plot Channels**. The GUI will display all currently excluded channels.

2. Edit Excluded Channels (Optional)

- In the plot, use each channel’s checkbox to include or exclude it manually.
- If you have made any changes, click **Save** to update the exclusion list.

3. Set Threshold

- Use the slider to adjust the detection threshold until it matches the noise level in your data.
- You can also:
 - Modify how many channels are shown at once.

- Adjust the amplitude scale for visibility.
- Type a numeric value directly into the threshold box.
- Note the threshold value, you will enter it in the Main tab before running the sorter.

3. Back to the Main Tab

1. Run and Sort

- In the **RUN** subtab, click **Run and Sort**.
- The software will:
 1. Extract samples.
 2. Run spike sorters on those samples and train classifiers.
 3. Apply the trained classifiers to the full dataset.
 - When sorting is finished, you will see a message saying:
“Sorting complete.”

4. Curation Tab

1. Load Sorted Data

- Click **Plot Sorting** to load the sorted output.
- If the dataset has been curated and saved before, the GUI restores the last saved curation automatically. **Reset** reverts to the raw sort.

2. Choose Plot Type

- The **Plot Type** dropdown lists single views (Amplitude, Waveform, CCG, ISI, Density, Amp Dist, PCs, Trace, Features) or **Multiple** to display several at once.
- **Amp Dist** shows a per-spike RMS distribution for every visible unit.
- The unit summary lives in its own window now. Click **Open table** to pop it out (see section 6). The ? button next to it opens the keyboard shortcut reference (see section 7).

3. Adjust Time Window

- Set the time window in the lower-right so plots focus on the period you care about.

4. Visualize Units

- Each unit row has two checkboxes. **Vis.** draws the unit's plots. **Select** highlights the unit so the Spike Group Controller acts on it.
- The unit list is paginated. Use the page buttons or type a page number to navigate.

5. Spike Group Controller

- **Edit selected units:**
 - **Remove** drops the selected units. Reversible with **Undo**.
 - **Merge** combines the selected units into the radio-button target.
 - **Split** clusters a unit's spikes using the **Method** (K-means default, GMM, or Graph) and **#Clust** fields. Splitting now uses the per-spike features the sorter saved, so cluster boundaries match the sorter's feature space.
 - **Limit** trims each selected unit to its longest stable interval. **Drop Rate** controls the change-point sensitivity.
 - **Isolation** cycles the SUA+ / SUA / MUA+ / MUA label.
 - **Undo** reverts the selected units. **Reset** reverts every edit since load.
 - **Notes** opens a small dialog to add a free-text note for the radio-selected unit. Notes are persisted with the saved curation. Inside the standalone curation table you can also double-click the Notes cell to edit in place.
 - **Realign** shifts the selected units' spike times by the slider value (ms).
- **Group Selection:**
 - Filter by **FR**, **ISI**, **SNR** or **Polarity**, then click **Include** or **Exclude** to tick the units that pass or fail.
 - Step through ticked units with the unit walker (Prev / Next + Unit #).
- **Similarity Processing:**
 - Click **Process** to compute pairwise waveform similarity. **Metric** picks the distance (XCorr, KL-div, or Bhattacharyya).
 - Use **Similarity**, **Amp. Var.** and **Dist (μ m)** to filter candidate pairs, walk them with the pair buttons, then merge via the Spike Group Controller.
- **Auto Curation:**
 - The **Auto Curation** panel runs a removal, cleaning, then merge pipeline. The gates are tuned from the panel:
 - **CCG** is the lag-zero peak / baseline-median ratio used by the cleaning phase. Higher is stricter. Default 1.5.
 - **Overlap** drops a unit if this fraction of its spikes coincide with a single neighbour.

- **Similarity**, **Amp. Var.** and **PC Dist** are the merge gates (waveform similarity, amplitude mismatch, normalised PC shape distance).
 - **ISI %** and **Budget** are the ISI gates for merging. **Dist (μm)** is the max y-distance considered between two units.
 - Click **Auto curation**. A summary dialog reports how many units were removed, cleaned and merged.
 - The **Pair type** dropdown walks what changed. Pair classes (Merged, Overlap removed) highlight two units at a time; unit classes (SUA+, SUA, MUA+, MUA, NaN) highlight one. The counter label switches between **#Pair** and **#Unit** accordingly.
- **Save:**
 - Click **Save** to write the curated sort, **curated_sample.mat** (which now also stores a session sub-struct so the next launch resumes the curation), and **curated_metrics.csv** into **<output>/RES_Sorted/**.
 - Closing the GUI also persists the panel state (gates, plot view, walker target) into the same folder, so re-opening the dataset comes back exactly where you left it.

6. Curation Table

- Click **Open table** (next to the plot dropdowns) to pop the per-unit summary into its own window. The window stays on top of the main GUI until you close it, and the column sort and scroll position survive across selection clicks.
- Columns: **G** (group index), **ID** (label after merges), **Ch** (main channel), **Rate (Hz)**, **ISI (%)**, **SNR**, **ISO** (isolation tier), **Stable** (% of recording), **Notes**. Numeric columns show two decimals and still sort numerically.
- Click any column header to sort by that column; click again to flip direction. **Refresh** rebuilds the table from the latest curation state (and resets the sort).
- Click any cell to set that row as the merge target (mirrors the radio button in the main GUI). Press **Space** to toggle Vis + Select on the focused row; selected rows are tinted with the units group colour.
- Removed units stay in the table in italic grey with **ISO = "REM"**. Bring them back with **Undo** (Ctrl+U) or **Reset** (Ctrl+R).
- The table window also carries Spike Group Controller buttons (Remove, Merge, Split, Limit, Isolation, Undo, Reset, Notes, Deselect) and a **Help** button with table-specific tips, so you can curate without switching back to the main figure.

7. Keyboard Shortcuts

- Click the **?** button next to **Open table** to see the full list. **Ctrl+?** opens the same dialog. All shortcuts use the Control key as the sole modifier (Mac users press the dedicated **control** / **^** key, separate from Cmd).

- Spike Group Controller: **Ctrl+D** Remove, **Ctrl+M** Merge, **Ctrl+T** Split, **Ctrl+L** Limit, **Ctrl+K** cycle Isolation, **Ctrl+U** Undo, **Ctrl+R** Reset, **Ctrl+E** edit Notes, **Ctrl+H** Deselect, **Ctrl+S** Save.
 - Navigation: **Ctrl+P** / **Ctrl+N** previous / next unit, **Ctrl+Up** / **Ctrl+Down** Auto walker, **Ctrl+[** / **Ctrl+]** Similarity walker, **Ctrl+Left** / **Ctrl+Right** previous / next page.
 - Inside the standalone curation table, **Space** toggles Vis + Select on the focused row, and every Ctrl+key shortcut above also works there.
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5. Help Tab

- For more information on any feature, click the **Help** tab. It provides full documentation on every tab, button, and parameter.